laboratory, whom many prefer not to see as they peer into their microscopes and fMRI scanners.[1]

In the eighteenth century *Homo sapiens* was like a mysterious black box, whose inner workings were beyond our grasp. Hence when scholars asked why a man drew a knife and stabbed another to death, an acceptable answer said: 'Because he chose to. He used his free will to choose murder, which is why he is fully responsible for his crime.' Over the last century, as scientists opened up the Sapiens black box, they discovered there neither soul, nor free will, nor 'self' – but only genes, hormones and neurons that obey the same physical and chemical laws governing the rest of reality. Today, when scholars ask why a man drew a knife and stabbed someone death, answering 'Because he chose to' doesn't cut the mustard. Instead, geneticists and brain scientists provide a much more detailed answer: 'He did it due to such-and-such electrochemical processes in the brain, which were shaped by a particular genetic make-up, which reflect ancient evolutionary pressures coupled with chance mutations.'

The electrochemical brain processes that result in murder are either deterministic or random or a combination of both – but they are never free. For example, when a neuron fires an electric charge, this may either be a deterministic reaction to external stimuli, or it might be the outcome of a random event such as the spontaneous decomposition of a radioactive atom. Neither option leaves any room for free will. Decisions reached through a chain reaction of biochemical events, each determined by a previous event, are certainly not free. Decisions resulting from random subatomic accidents aren't free either. They are just random. And when random accidents combine with deterministic processes, we get probabilistic outcomes, but this too doesn't amount to freedom.

Suppose we build a robot whose central processing unit is linked to a radioactive lump of uranium. When choosing between two options – say, press the right button or the left button – the robot counts the number of uranium atoms that decayed during the previous minute. If the number is even – it presses the right button. If the number is odd – the left button. We can never be certain about the actions of such a robot. But nobody would call